

Predictive Model Plan

Customer Delinquency Prediction – Geldium
Tata GenAI Powered Data Analytics Job Simulation – Task 2

1. Model Logic (Generated with GenAI)

The model is designed to predict whether a customer will become **delinquent** on their account (**Delinquent_Account** = 1) or remain in good standing (**Delinquent_Account** = 0), using their financial and behavioural attributes such as income, credit score, credit utilization, missed payments, debt-to-income ratio, employment status and their payment record over the last six months. Below is the step-by-step logic of the model, along with the corresponding implementation.

Step-by-step process

- **Import libraries** - load pandas, numpy and the required scikit-learn modules (train_test_split, LogisticRegression, accuracy_score, classification_report).
- **Load the data** - read the customer dataset (Delinquency_prediction_dataset.xlsx) into a DataFrame and inspect it using df.head(), df.tail(), df.info(), df.describe() and df.shape.
- **Check data quality** - use df.isnull().sum() to identify missing values in Income, Credit_Score and Loan_Balance.
- **Clean the data** - fill missing Income with the column mean; fill missing Credit_Score and Loan_Balance with the column median; drop the non-predictive Customer_ID column.
- **Define features and target** - set y = Delinquent_Account (the label to predict) and x = all remaining columns (the predictors).
- **Encode categorical variables** - convert text columns (Employment_Status, Location, Credit_Card_Type, Month_1-Month_6) into numeric form using one-hot encoding (pd.get_dummies).
- **Split the data** - divide x and y into training (80%) and test (20%) sets using train_test_split with random_state = 42 for reproducibility.
- **Train the model** - fit a Logistic Regression classifier on the training data (x_train, y_train) with max_iter = 1000.
- **Generate predictions** - use the trained model to predict delinquency on the unseen test set (x_test), producing y_pred.
- **Evaluate performance** - compare y_pred against the actual y_test values using accuracy_score to measure how well the model performs.

Simplified code snippet

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

df = pd.read_excel("Delinquency_prediction_dataset.xlsx")
```

```
df["Income"].fillna(df["Income"].mean(), inplace=True)
df["Credit_Score"].fillna(df["Credit_Score"].median(), inplace=True)
df["Loan_Balance"].fillna(df["Loan_Balance"].median(), inplace=True)

y = df["Delinquent_Account"]
x = df.drop(["Delinquent_Account", "Customer_ID"], axis=1)
x = pd.get_dummies(x, drop_first=True)

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42)

model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)

y_pred = model.predict(x_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy of the model is:", accuracy) # 0.86
```

2. Justification for Model Choice

Logistic Regression was selected as the model for this delinquency prediction task for the following reasons:

- **Accuracy** - On this dataset, the model achieved **86% accuracy** on the held-out test set, which is a solid baseline result and noticeably better than a naive rule that always predicts "not delinquent" (which would be correct only around 84% of the time, given the ~16% delinquency rate in the data).
- **Transparency** - Logistic Regression is a "white-box" model: each feature gets a coefficient that directly shows the direction and relative strength of its effect on delinquency risk (e.g. higher Missed_Payments increases risk, higher Credit_Score decreases it). This makes it far easier to explain predictions to non-technical stakeholders than a "black-box" model like a neural network.
- **Ease of use / implementation** - It required minimal preprocessing beyond handling missing values and one-hot encoding categorical variables, trains quickly even on modest hardware, and has very few hyperparameters to tune, making it practical to implement and maintain.
- **Relevance for financial prediction** - Logistic Regression is an industry-standard technique for credit scoring and delinquency/default prediction because it directly models the probability of a binary outcome (delinquent vs. not delinquent), which financial institutions can use as a risk score rather than just a hard 0/1 label.
- **Suitability for Geldium's business needs** - Since Geldium's risk and collections teams need to understand *why* a customer is flagged as high-risk (not just receive a prediction), an interpretable model like Logistic Regression fits their need for explainable, auditable decisions - an important consideration in regulated financial environments where lending decisions may need to be justified to customers or regulators.

As a next iteration, tree-based ensemble models (e.g. Random Forest, Gradient Boosting) could be explored to check whether they capture non-linear relationships better, but they would trade off some of the interpretability that Logistic Regression offers.

3. Evaluation Strategy

The current model was evaluated using accuracy on a held-out 20% test set. To fully validate the model before deployment, the following evaluation strategy is recommended:

Metrics to use

- **Accuracy** - overall proportion of correct predictions; currently **86%**. Useful as a headline number but can be misleading on imbalanced data.
- **Precision** - of all customers predicted delinquent, what fraction actually were; important to avoid unnecessarily flagging good customers for restrictive action.
- **Recall (Sensitivity)** - of all customers who were actually delinquent, what fraction the model correctly caught; critical here, since missing a genuinely at-risk customer is typically more costly for a lender than a false alarm.
- **F1-score** - the harmonic mean of precision and recall, giving a single balanced measure - especially useful since only ~16% of customers in the dataset are delinquent (a class-imbalanced problem).
- **AUC-ROC** - measures how well the model separates delinquent from non-delinquent customers across all classification thresholds, which is useful since the business may want to adjust the risk threshold rather than use a fixed 0.5 cut-off.
- **Confusion matrix** - to see the exact breakdown of true positives, false positives, true negatives and false negatives, which grounds all the above metrics in concrete numbers.

Interpreting the metrics

A high accuracy alone would not be trusted given the class imbalance; precision and recall for the "delinquent" (minority) class would be reviewed together. If recall is low, the model is missing at-risk customers and the classification threshold could be lowered (accepting more false positives) to catch more true delinquents, since the cost of missing a risky customer is generally higher than the cost of extra monitoring on a safe one. If precision is low, too many good customers are being flagged, which could hurt customer experience and operational efficiency in collections.

Bias detection and mitigation

- Check model performance separately across subgroups such as Location and Employment_Status to confirm the model is not systematically over- or under-predicting delinquency for any particular group.
- Review whether any feature acts as a proxy for a protected characteristic (e.g. certain locations correlating with income level) and assess whether it should be reweighted, transformed or excluded.
- Re-check the class balance regularly - since only ~16% of the training data is delinquent, the model could be retrained with techniques like class weighting or resampling if it is found to be biased toward predicting "not delinquent".
- Monitor performance over time on new data, since customer behaviour and economic conditions change, and a model that was fair and accurate at training time can drift.

Ethical considerations

- **Transparency to customers** - customers flagged as high-risk should be able to understand, at a high level, why the account was flagged, especially if the prediction affects their credit limit or loan terms.
- **Avoiding unfair harm** - predictions should support early, supportive interventions (reminders, flexible repayment plans) rather than be used punitively in ways that could push a borderline customer further into financial difficulty.

- **Data privacy** - customer financial and behavioural data must be handled securely and used only for its intended risk-assessment purpose, in line with data protection regulations.
- **Human oversight** - the model's output should support, not replace, human judgement in high-stakes decisions such as denying credit or closing an account.